

Developer Mock Test Results

Test ID:	2428c427-eeaa-4b...	Student ID:	112
Test Type:	Developer Assessment	Date:	2026-03-05 10:36:28
Overall Score:	10/25 (40.0%)	Performance:	■ Average
Warnings:	0/3	Status:	Completed ■

Section-wise Performance

Section	Score	Percentage	Status
■ Aptitude	3/10	30.0%	■■ Needs Work
■ MCQ/Theory	3/10	30.0%	■■ Needs Work
■ Coding	4/5	80.0%	■ Pass

Detailed Question Review

■ APTITUDE SECTION (3/10 - 30.0%)

■ Q1. What is the next number: 1, 3, 5, 7, 9, ?

Your Answer: 11

■ Well done!

■ Q2. If it is true that all cars are vehicles and some vehicles are bicycles, what can be concluded?

Your Answer: Some cars are bicycles

Correct Answer: Some vehicles are cars

■ The correct answer is "Some vehicles are cars" because we know that all cars are vehicles, which implies that the category of cars is a subset of the category of vehicles. The user's answer, "Some cars are bicycles", is incorrect because the statement "some vehicles are bicycles" does not necessarily imply that those bicycles are also cars. The user likely mistakenly assumed an overlap between the categories of cars and bicycles based on their shared relationship with vehicles.

■ Q3. A mixture is made of 3 parts sugar and 2 parts water. What is the ratio of sugar to water?

Your Answer: 1:1

Correct Answer: 3:2

■ The correct answer is 3:2 because the mixture is made of 3 parts sugar and 2 parts water, directly giving us the ratio of sugar to water. The key calculation step is simply comparing the parts of sugar to the parts of water: 3 parts sugar / 2 parts water = 3:2 ratio. The user's likely mistake was misinterpreting the given information, as they incorrectly stated the ratio as 1:1, which would imply equal parts of sugar and water, not 3 parts sugar to 2 parts water.

■ Q4. What is the next number: 1, 4, 9, 16, 25, ?

Your Answer: 42

Correct Answer: 36

■ The correct answer is 36 because the pattern of the sequence is obtained by squaring consecutive integers: $1^2 = 1$, $2^2 = 4$, $3^2 = 9$, $4^2 = 16$, $5^2 = 25$, and $6^2 = 36$. The user's answer, 42, is likely incorrect because it doesn't follow this pattern of squaring consecutive integers. The key calculation step to determine the next number in the sequence is $6^2 = 36$, which confirms that the correct answer is indeed 36.

■ **Q5. If 5 workers can build a house in 12 days, how many days will it take for 3 workers to build the same house?**

Your Answer: 15 days

Correct Answer: 20 days

■ To find the number of days it will take for 3 workers to build the house, we can use the formula: (number of workers initial) * (number of days initial) = (number of workers final) * (number of days final). So, 5 workers * 12 days = 3 workers * x days, which gives us $5 * 12 = 3 * x$, then $60 = 3x$, and finally $x = 60 / 3 = 20$ days. The user's mistake was likely in not using this formula or not calculating the result correctly, as the correct calculation shows it will take 20 days for 3 workers to build the house.

■ **Q6. If it is true that all students are learners and some learners are teachers, what can be concluded?**

Your Answer: Some students are teachers

Correct Answer: Some learners are students

■ The correct answer is "Some learners are students" because we know that all students are learners, which implies that the set of students is a subset of the set of learners. The user's answer, "Some students are teachers", is incorrect because it assumes a relationship between students and teachers that is not necessarily supported by the given information. The user likely made a mistake by reversing the relationship between learners and students, and then applying it to the relationship between learners and teachers.

■ **Q7. What is the average of 2, 4, 6, 8, 10?**

Your Answer: 6

■ Correct! Well done.

■ **Q8. If 3 men can build a wall in 8 days, how many men are needed to build the same wall in 4 days?**

Your Answer: 4

Correct Answer: 6

■ To find the number of men needed to build the wall in 4 days, we can use the formula: (number of men) * (number of days) = constant. Given that 3 men can build the wall in 8 days, we have $3 * 8 = 24$. To build the same wall in 4 days, we set up the equation: (number of men) * 4 = 24. Solving for the number of men gives: number of men = $24 / 4 = 6$. The user's answer of 4 is likely incorrect because it doesn't account for the proportional increase in manpower needed to halve the construction time.

■ **Q9. The average of 6 numbers is 18. If one of the numbers is 20, what is the sum of the remaining 5 numbers?**

Your Answer: 100

■ Correct! Well done.

■ **Q10. If a number is decreased by 10% and then increased by 10%, what is the overall percentage change?**

Your Answer: 0%

Correct Answer: -1%

■ The correct answer is -1% because when a number is decreased by 10% and then increased by 10%, the overall effect is not a complete reversal to the original value. The key calculation step is: if we start with 100, decreasing it by 10% gives 90, and then increasing 90 by 10% gives 99, which is 1 less than the original 100, resulting in a -1% overall change. The user's likely mistake was assuming that the decrease and increase would cancel each other out, which is not the case due to the percentage changes being applied to different bases (100 and 90).

■ **MCQ SECTION (3/10 - 30.0%)**

■ Q1. How can a Java program be executed?

Your Answer: Using an IDE

■ Excellent! Right answer.

■ Q2. What is a RuntimeException in Java?

Your Answer:

A type of error that is thrown at runtime

Correct Answer:

A type of exception that is thrown at runtime

■ The correct answer is A) A type of exception that is thrown at runtime, because a RuntimeException in Java is a subclass of the Exception class that is thrown during the execution of a program, indicating an error that can be recovered from. The user's answer, "A type of error that is thrown at runtime", is incorrect because it confuses exceptions and errors - RuntimeException is a type of exception, not error. The user's likely mistake is not understanding the distinction between exceptions and errors in Java, where exceptions are instances of the Exception class and errors are instances of the Error class.

■ Q3. What helps developers fix problems quickly in Java?

Your Answer: Error messages

■ Well done!

■ Q4. What is the basic syntax of Java?

Your Answer: Variables, data types, control structures, and loops

Correct Answer: Variables, data types, control structures, methods, classes, and objects

■ The correct answer is A) Variables, data types, control structures, methods, classes, and objects because it encompasses the fundamental components of Java syntax. The user's answer, D) Variables, data types, control structures, and loops, is incorrect because it misses key elements such as methods, classes, and objects, which are essential to Java's object-oriented programming paradigm. The user's likely mistake is oversimplifying the syntax of Java, failing to account for its object-oriented nature and the importance of methods and classes in the language.

■ Q5. What are packages in Java?

Your Answer: Collections of related methods

Correct Answer: Collections of related classes

■ The correct answer is "Collections of related classes" because in Java, packages are used to organize related classes and interfaces that are used for a specific purpose or functionality. The user's answer, "Collections of related methods", is incorrect because methods are part of classes, and packages group classes, not individual methods. The user's mistake is likely due to confusing the purpose of packages with the purpose of classes, which can contain multiple related methods.

■ Q6. What is an Error in Java?

Your Answer:

A type of exception that is thrown at compile-time

Correct Answer:

A type of exception that is not recoverable

■ The correct answer is C) A type of exception that is not recoverable, because in Java, an Error is a subclass of Throwable that is thrown when a serious, unrecoverable error occurs, such as an OutOfMemoryError. The user's answer, B) A type of exception that is thrown at compile-time, is incorrect because Errors are typically thrown at runtime, not compile-time. The user's mistake likely stems from confusing Errors with compile-time errors, which are syntax errors that prevent the code from compiling.

■ Q7. How does Java provide error handling?

Your Answer: Using if-else statements

Correct Answer: Using try, catch, and finally blocks

■ The correct answer, "Using try, catch, and finally blocks", is right because Java provides a built-in mechanism for error handling through these blocks, allowing developers to enclose code that may throw exceptions, catch and handle

exceptions, and execute final code regardless of exceptions. The user's answer, "Using if-else statements", is incorrect because while if-else statements can be used for conditional checks, they are not the primary mechanism for error handling in Java. The user likely mistook conditional statements for error handling, which are two distinct concepts in programming.

■ Q8. What type of applications can be built using Java?

Your Answer: Web development, mobile apps

■ Correct! Well done.

■ Q9. What is optimization in Java?

Your Answer:

The process of improving the functionality of a program

Correct Answer:

The process of improving the performance of a program

■ The correct answer is "The process of improving the performance of a program" because optimization in Java refers to the process of modifying a program to make it more efficient, reducing its execution time, and improving its overall performance. The user's answer, "The process of improving the functionality of a program", is incorrect because it describes the process of adding new features or enhancing existing ones, rather than improving performance. The user's mistake likely stems from confusing optimization with enhancement or feature development, which are distinct concepts in software development.

■ Q10. What is required to run Java programs?

Your Answer: Java JRE

Correct Answer: Java JDK

■ The correct answer is Java JDK because it includes the Java Runtime Environment (JRE) and development tools, such as a compiler, which are necessary to run Java programs. The user's answer, Java JRE, is incorrect because while it can run Java programs, it does not include the development tools needed to compile them. The user likely mistook the JRE, which is a subset of the JDK, as the primary requirement to run Java programs, when in fact the JDK is the more comprehensive package that includes the JRE.

■ CODING SECTION (4/5 - 80.0%)

■ Q1. Write a Java program that reads a string from stdin and prints its reverse. ****Input Format:**** - Line 1: A string ****Output Format:**** - The reversed s...

Your Answer:

```
import java.util.Scanner;

public class Main {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        String input = sc.nextLine();
        String reversed = "";

        for (int i = input.length() - 1; i >= 0; i--) {
            reversed += input.charAt(i);
        }

        System.out.println(reversed);

        sc.close();
    }
}
```

■ Your code has successfully passed all test cases, demonstrating a thorough understanding of the problem and its solution. Great job on achieving an "Accepted" status, indicating that your code is correct and effective.

■ Q2. Write a program that reads a list of integers and prints the sorted list. ****Input Format:**** - Line 1: Number of integers (integer) - Line 2: List of ...

Your Answer:

```
import java.util.*;

public class Main {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        int n = sc.nextInt();
        int[] arr = new int[n];

        for(int i = 0; i < n; i++) {
            arr[i] = sc.nextInt();
        }

        Arrays.sort(arr);

        for(int i = 0; i < n; i++) {
            System.out.print(arr[i] + " ");
        }

        sc.close();
    }
}
```

■ Your code has successfully passed all test cases, demonstrating a thorough understanding of the problem and its solution. Great job on achieving an "Accepted" status, indicating that your code is correct and efficient.

■ Q3. Write a Java program that reads an array of integers from stdin and finds the minimum and maximum values. ****Input Format:**** - Line 1: The number of e...

Your Answer:

```
import java.util.Scanner;

public class Main {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        int n = sc.nextInt();
        int[] arr = new int[n];

        for (int i = 0; i < n; i++) {
            arr[i] = sc.nextInt();
        }

        int min = arr[0];
        int max = arr[0];

        for (int i = 1; i < n; i++) {
            if (arr[i] < min) {
                min = arr[i];
            }
            if (arr[i] > max) {
                max = arr[i];
            }
        }

        System.out.println(min + " " + max);

        sc.close();
    }
}
```

```
}
```

■ Your code has successfully passed all test cases, indicating a correct and effective implementation of the solution. Great job on achieving an "Accepted" status, keep up the good work!

■ **Q4. Write a Java program that reads an integer and a precision from stdin and checks if the number is a perfect cube with the given precision. **Input Fo...**

Your Answer:

```
import java.util.Scanner;

public class Main {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        int num = sc.nextInt();
        double precision = sc.nextDouble();

        double cubeRoot = Math.cbrt(num);
        double nearestInt = Math.round(cubeRoot);

        if (Math.abs(cubeRoot - nearestInt) <= precision) {
            System.out.println("Perfect Cube");
        } else {
            System.out.println("Not Perfect Cube");
        }

        sc.close();
    }
}
```

■ Your code has successfully passed all test cases, indicating a correct and effective implementation of the solution. Great job on achieving an "Accepted" status, keep up the good work!

■ **Q5. Write a Java program that reads two arrays of integers from stdin and merges them into a**

Your Answer:

```
import java.util.Scanner;

public class Main {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        int n1 = sc.nextInt();
        int[] arr1 = new int[n1];

        for (int i = 0; i < n1; i++) {
            arr1[i] = sc.nextInt();
        }

        int n2 = sc.nextInt();
        int[] arr2 = new int[n2];

        for (int i = 0; i < n2; i++) {
            arr2[i] = sc.nextInt();
        }

        int[] merged = new int[n1 + n2];

        for (int i = 0; i < n1; i++) {
            merged[i] = arr1[i];
        }
    }
}
```

```

        for (int i = 0; i < n2; i++) {
            merged[n1 + i] = arr2[i];
        }

        for (int num : merged) {
            System.out.print(num + " ");
        }

        sc.close();
    }
}

```

Correct Answer:

```

import java.util.Scanner;

public class Main {
    public static void main(String[] args) {
        Scanner scanner = new Scanner(System.in);

        System.out.println("Enter the size of the first array:");
        int size1 = scanner.nextInt();
        int[] array1 = new int[size1];

        System.out.println("Enter the elements of the first array:");
        for (int i = 0; i < size1; i++) {
            array1[i] = scanner.nextInt();
        }

        System.out.println("Enter the size of the second array:");
        int size2 = scanner.nextInt();
        int[] array2 = new int[size2];

        System.out.println("Enter the elements of the second array:");
        for (int i = 0; i < size2; i++) {
            array2[i] = scanner.nextInt();
        }

        int[] mergedArray = new int[size1 + size2];
        System.arraycopy(array1, 0, mergedArray, 0, size1);
        System.arraycopy(array2, 0, mergedArray, size1, size2);

        System.out.println("Merged array:");
        for (int i = 0; i < mergedArray.length; i++) {
            System.out.print(mergedArray[i] + " ");
        }

        scanner.close();
    }
}

```

■ The student's code failed because it appears to be skipping the first element of the first array and the first element of the second array is also not being read correctly. The actual output '2 4 5 6' suggests that the code is starting from the second element of the first array and the second element of the second array. This indicates an issue with how the input is being read or processed.

■ Recommendations

Areas that need improvement:

- **Aptitude:** Practice more logical reasoning, quantitative aptitude, and problem-solving questions.
- **MCQ/Theory:** Review programming concepts, data structures, and algorithms theory.